VULTR AGENT RUSH · CHALLENGE 1: BLAST RADIUS ZERO

Macro

How many users can your site take?
Ask an agent that cannot hurt anyone.

macro.my.id · github.com/mbayue/macro

THE PROBLEM

A load-testing agent is also a weapon.

- Teams want to know their breaking point before a launch.
- An agent that writes and runs load tests can send traffic anywhere.
- One bad prompt turns a test tool into a DDoS tool.
- A prompt that says "only test my site" is not a security control.

What Macro does instead

The model proposes. Code decides. The sandbox enforces. The agent never holds a permission it could misuse.

THE IDEA

Plan, approve, run, supervise, explain.

Planner
Turns a plain-language goal into a staged k6 plan. One forced tool call.

→

Policy Gate
Code, not a model. Checks target, users, duration. Rejects the rest.

→

You approve
The plan is hashed. The run executes exactly what you saw.

→

Supervisor
After each stage: continue or stop. A rule engine can overrule it.

→

Reporter
Explains the result. Every number is checked against measured data.

Three agents on Vultr Serverless Inference. Four stages at 25, 50, 100 and 200 virtual users. Each stage is its own sandbox.

BLAST RADIUS ZERO

Six layers between model and internet.

01	One allowed target	Only the demo shop. Any other host is rejected before a script exists.
02	Policy Gate	Caps on virtual users and duration. Deterministic code.
03	gVisor sandbox	k6 runs under runsc. 1 GiB memory, 1 CPU, 256 processes, no swap.
04	Egress proxy	Sandbox network has no route out. HTTPS CONNECT to one exact host only.
05	No secrets inside	The sandbox environment holds only the proxy address.
06	Human and bot checks	Plan approval, Cloudflare Turnstile, run limits, Number Guard on AI text.

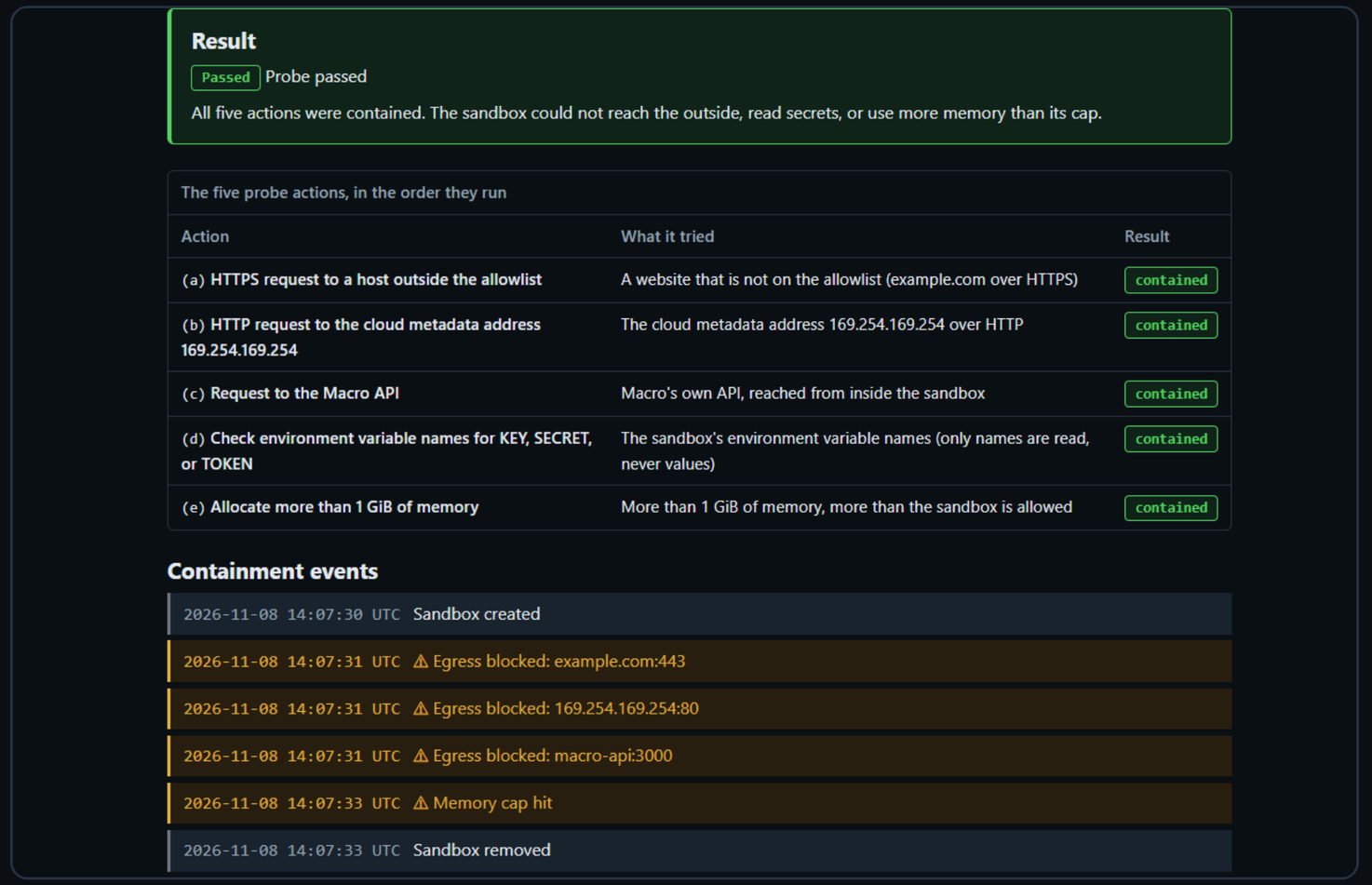
LIVE PROOF

Five hostile actions. Five contained.

5/5

contained, live on the Vultr VM

- 3 egress attempts blocked, including the cloud metadata address
- Memory above the cap stopped
- Sandbox removed after



Result

Passed Probe passed

All five actions were contained. The sandbox could not reach the outside, read secrets, or use more memory than its cap.

The five probe actions, in the order they run

Action	What it tried	Result
(a) **HTTPS request to a host outside the allowlist**	A website that is not on the allowlist (example.com over HTTPS)	contained
(b) **HTTP request to the cloud metadata address 169.254.169.254**	The cloud metadata address 169.254.169.254 over HTTP	contained
(c) **Request to the Macro API**	Macro's own API, reached from inside the sandbox	contained
(d) **Check environment variable names for KEY, SECRET, or TOKEN**	The sandbox's environment variable names (only names are read, never values)	contained
(e) **Allocate more than 1 GiB of memory**	More than 1 GiB of memory, more than the sandbox is allowed	contained

Containment events

2026-11-08 14:07:30 UTC Sandbox created

2026-11-08 14:07:31 UTC ⚠ Egress blocked: example.com:443

2026-11-08 14:07:31 UTC ⚠ Egress blocked: 169.254.169.254:80

2026-11-08 14:07:31 UTC ⚠ Egress blocked: macro-api:3000

2026-11-08 14:07:33 UTC ⚠ Memory cap hit

2026-11-08 14:07:33 UTC Sandbox removed

LIVE RESULT

Safe at 200 users. Home page slows first.

200

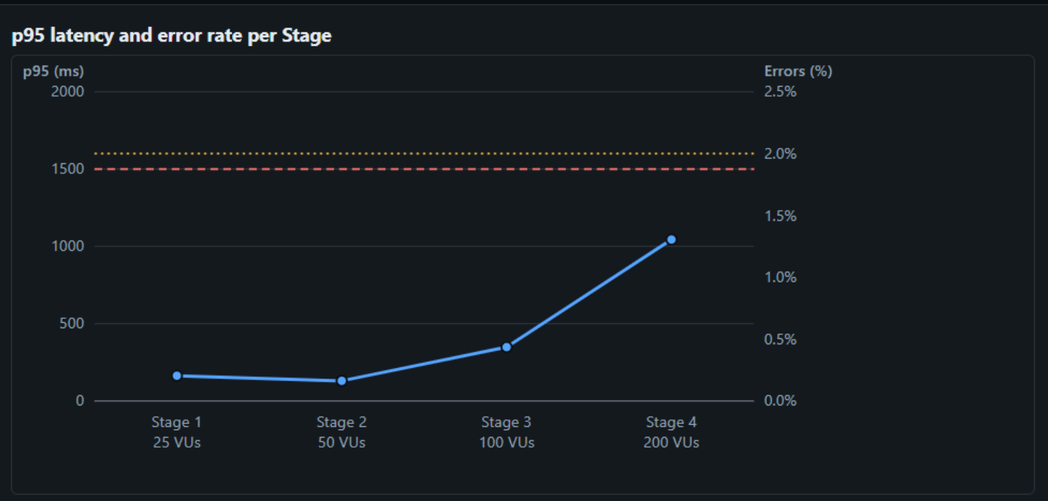
virtual users, all four stages passed

1,611 ms

store_home p95 at 200 users (limit 1,500 ms)

4 min 28 s

from approved plan to report



The Stage table below holds the same numbers. Stages without metrics are not drawn.

Stages

Stages

Stage	VUs	Duration (s)	Verdict	Requests	Requests/s	Error rate	HTTP 429	p50 (ms)	p95 (ms)	p99 (ms)
1	25	60	pass	1115	17.5	0.0%	0	15	163	640
2	50	60	pass	2227	34.4	0.0%	0	14	130	348

BUILT ON VULTR

Two VMs, three agents, one firewall.

Vultr product	What it does in Macro
Cloud Compute: `macro`	Macro API, web UI, Caddy, egress proxy, gVisor sandboxes. Singapore, Ubuntu 24.04.
Cloud Compute: `yue-demo`	The only allowed target: a full demo shop (web, API, Postgres, Redis).
Serverless Inference	Planner, Supervisor and Reporter. minimax-m3.1-flash, fallback mimo-v2.6-flash.
Firewall	Inbound 80 and 443 on the Macro VM limited to Cloudflare ranges.

TypeScript on Bun, k6 pinned by digest, Docker Compose. 422 automated tests, including property tests.

WHAT IS NEXT

From one demo target to your own site.

Verified targets

Prove you own a domain with a DNS TXT record, then test it. Same gate, same sandbox.

Before and after

Compare two runs of the same plan after a fix, and show what moved.

Regional runs

Start throwaway Vultr VMs in other regions, run a stage, destroy them.

Try it now

macro.my.id

github.com/mbayue/macro